UNIVERSAL CLOSEDNESS OF A REPRESENTABLE MORPHISM OF ALGEBRAIC SPACES TESTED BY BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof shares its mechanism with the submersiveness and openness criteria but is written out separately by the author; the closed case is counted once here under its own claim_key.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Description of the points of a fibre product of algebraic spaces. Properties of Spaces, Section spaces-properties-section-points, used at proof lines 1532-1543. Gives the commutativity of the square and the openness, surjectivity and comparison surjectivity used.
- Universal closedness of a representable morphism in the functorial sense. Section section-representable, used at proof lines 1518-1522 and 1529-1531. Fixes condition (1) and gives closedness of V x_Y X -> V.
- Closedness of a map detected after a surjective open comparison. used at proof lines 1540-1546. Transfers closedness from the left to the right vertical arrow.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03BT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4360–4360. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a representable morphism of algebraic spaces over S. The following are equivalent*

(1) *f is universally closed (in the sense of Section 03HA), and*
(2) *for every morphism of algebraic spaces $Z \to Y$ the morphism of topological spaces $|Z \times_Y X| \to |Z|$ is closed.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Assume (1), and let $Z \to Y$ be as in (2). Choose a scheme V and a surjective étale morphism $V \to Y$. By assumption the morphism of schemes $V \times_Y X \to V$ is universally closed. By Properties of Spaces, Section 03BT in the commutative diagram

$$\begin{array}{ccc} |V \times_Y X| & \longrightarrow & |Z \times_Y X| \\ \downarrow & & \downarrow \\ |V| & \longrightarrow & |Z| \end{array}$$

the horizontal arrows are open and surjective, and moreover

$$|V \times_Y X| \longrightarrow |V| \times_{|Z|} |Z \times_Y X|$$

is surjective. Hence as the left vertical arrow is closed it follows that the right vertical arrow is closed. This proves (2). The implication (2) $\Rightarrow$ (1) is immediate from the definitions. $\square$